

## No. of Functional Features Included in the Solution

|               |                                                              |
|---------------|--------------------------------------------------------------|
| Date          | 15 March 2024                                                |
| Team ID       | XXXXXX                                                       |
| Project Name  | OptiCrop : Smart Agricultural Production Optimization Engine |
| Maximum Marks | 5 Marks                                                      |

### Functional Features Overview:

This document summarizes the functional features implemented in the OptiCrop system. Each feature contributes to accurate crop recommendation, improved user interaction, and efficient machine learning-based prediction.

| S.No | Feature Name       | Feature Description               | Module / Component | Status (Done / In Progress / Pending) | Marks Contribution |
|------|--------------------|-----------------------------------|--------------------|---------------------------------------|--------------------|
| 1    | Crop Suggestion    | Predicts suitable crops using ML. | Prediction         | Done                                  | 1                  |
| 2    | Data Preprocessing | Cleans and prepares dataset.      | Preprocessing      | Done                                  | 1                  |
| 3    | Model Training     | Trains ML model for prediction.   | ML Model           | Done                                  | 1                  |
| 4    | Flask Web App      | Interactive web interface.        | Frontend           | Done                                  | 1                  |
| 5    | User Input Form    | Accepts soil and climate input    | UI                 | Done                                  | 0.5                |
| 6    | Prediction Output  | Displays recommended crop.        | Prediction         | Done                                  | 0.5                |
| 7    | Data Visualization | Visualizes dataset insights.      | EDA                | Done                                  | 0.5                |
| 8    | Model Evaluation   | Measures prediction accuracy.     | Testing            | Done                                  | 0.5                |

### Feature Summary:

| Metric                     | Count / Value |
|----------------------------|---------------|
| Total Features Planned     | 8             |
| Total Features Implemented | 8             |
| Core / Must-Have Features  | 5             |